केस स्टडी/ CASE STUDY

OISD/CS/2026-27/P&E/10 **Date: 06.08.2026**

INTRODUCTION

Title: Fatal flash fire during opening of an isolated desalter manway (CDU)
Location: Refinery
Loss/ Outcome: Fatal injury

BRIEF OF INCIDENT

Following a major Maintenance & Inspection (M&I) shutdown, Crude Distillation Unit (CDU) was restarted. During the shutdown, inspection and maintenance of the first-stage desalter, second-stage desalter and their associated transformer systems were carried out. After startup, drawing of high current was observed in the transformer associated with the first-stage desalter. Despite troubleshooting, abnormal high current draw persisted. Hence, it was decided to isolate and handover the desalter for internal inspection.

The desalter was bypassed, isolated and depressurized. Demineralized (DM) water was introduced from bottom into the vessel to displace the hydrocarbon inventory through desalter outlet header. After displacement of crude oil through the top of the desalter, water wash was carried out. Blinding activities were eventually completed, except for the LP steam line, which remained unblinded as steaming was planned later.

On the day of the incident, after completion of water washing, work permits were issued for de-torquing and opening of the desalter manways. Contractor manpower was instructed to open both the north-side and south-side manways of the desalter. Two contract workmen (hereinafter referred to as IP-1 and IP-2) were engaged in opening the south-side manway. Immediately after removal of the last remaining bolt and opening of the manway cover, a flash fire accompanied by blast occurred. IP-1 fell from the working platform (at a height of approximately 3.5 m) and sustained fatal injuries.

IP-2 climbed down from the platform using the monkey ladder. No perceptible injury except minor burn injuries in face was reported.

The Manual Call Point (MCP) was activated, and emergency response personnel, fire tenders and an ambulance were mobilized. Smoke continued to emerge from the desalter, and fire water was applied into the vessel to cool the equipment and suppress the residual fire and smoke.

Both IP-1 and IP-2 were shifted to the Occupational Health Centre (OHC) after the incident. IP-1 was provided necessary medical attention; however, no signs of recovery were observed. Owing to the severity of his injuries, IP1 was transported by ambulance to the City Hospital for further treatment, where he was declared dead.

Although IP-2 initially remained conscious, oriented and clinically stable under medical observation, his condition deteriorated approximately four hours later with complaints of giddiness and mild abdominal discomfort. He was subsequently shifted to the City Hospital for further evaluation and treatment. Despite medical intervention, he succumbed later on the same day.

OBSERVATIONS/ SHORTCOMINGS

1. Blinding activities commenced while vessel preparation and secondary water washing were still in progress, indicating inadequate sequencing and verification during equipment handover.
2. The spectacle blind on the LP steam line was maintained in the through (open) position instead of the designated blind position specified in the approved P&ID, indicating deviation from the approved process configuration and deficiencies in equipment isolation and handover practices. (Refer- Figure:1)
3. Evidence could not establish that mandatory decontamination, steaming, hot water washing, and verification of zero hydrocarbon/ toxic gas conditions were completed prior to opening the vessel, contrary to approved procedures.
4. Accumulation of sludge inside the desalter was observed after the incident thereby implying the possibility of flammable hydrocarbon atmosphere during vessel opening.
5. Transformer testing/ checking activities were reportedly undertaken after electrical isolation without evidence of corresponding LOTO implementation, indicating a possible bypass of the electrical isolation system. The possibility of energized desalter grids at the time of vessel opening could not be ruled out.
6. A flash fire occurred immediately upon opening of the desalter manway, resulting in fatal injuries to two workmen.
7. DCS trends showed a sudden pressure spike and simultaneous temperature rise inside the desalter coinciding with the incident, corroborating the occurrence of a flash fire accompanied by pressure wave. (Refer- Figure:2)
8. One injured workman initially appeared stable with only minor external burns but later succumbed due to internal injuries, highlighting the need for immediate referral of personnel exposed to blast/ flash fire incidents to advanced trauma care facilities.
9. Damage to the platform safety railing and surrounding structures indicated that the flash fire was accompanied by a blast (pressure wave). (Refer- Figure:3)
10. The burn injuries, fall from height, and post-mortem findings were consistent with simultaneous exposure to flash fire and pressure wave effects.
11. The crude outlet piping configuration lacked adequate low-point drainage provisions, creating the possibility of hydrocarbon entrapment and subsequent migration back into the desalter during depressurization and draining. (Refer- Figure:4)
12. Significant gaps were observed in permit implementation, including inadequate identification of job hazards, absence of documented electrical isolation/ LOTO verification, omission of gas testing requirements, and incomplete verification of equipment readiness prior to authorizing maintenance activities.

PROBABLE REASONS OF FAILURE / ROOT CAUSE

Source of Hydrocarbon

The source of hydrocarbon was due to inadequate flushing and purging activities.

Probable source of ignition

1. The ignition may have been due to spontaneous autoignition of pyrophoric iron sulphide deposits coming in contact with air during opening the manhole cover. Though the vessel was in operation only for few days after the M&I, sludge presence after the incident indicates probable presence of pyrophoric iron.
2. Possible non-compliance with LOTO requirements during transformer testing/checking activities could not be ruled out.

3. Generation of an ignition spark due to metal-to-metal contact during removal of the final studs/nuts and cover opening can also be a probable ignition trigger.

RECOMMENDATIONS

1. Approved SOPs/ Manual shall be followed during equipment handover and maintenance preparation activities. Any deviation from the approved procedures shall be undertaken only through a formal authorization and defined approval process, including Management of Change (MOC), in accordance with Clause 7.2.5 (d) of OISD-GDN-206 and Clause 4.2 of OISD-STD-178.

   The task team should be aware of the SOP and associated risk with mitigations in place before starting the job as per Clause 7.2.4(h) of OISD-GDN-206.

2. Strict compliance to Electrical Lock-out and Tag-out (LOTO) procedure shall be ensured for all activities involving testing (including temporary testing), checking, switching, energization, de-energization, maintenance, or troubleshooting of electrical equipment.

   In addition to tag-out in sub-station, display of the LOTO tags at site shall also be implemented to make all the concerned aware of the isolation. The adequacy of the existing LOTO procedure shall be reviewed and strengthened.

3. Opening of process equipment, where pyrophoric iron sulphide (FeS) deposits may be present, shall be carried out in accordance with the safety precautions specified in the applicable licensor guidelines, approved operating manuals, and relevant procedures. Where such hazards and associated control measures are not adequately addressed in the existing SOPs or operating manuals, the same shall be incorporated based on licensor recommendations and recognized good engineering practices.

4. Prior to blinding and subsequent opening of process equipment for maintenance, complete evacuation and verification of hydrocarbon/ toxic gases shall be established including in the connected piping/ header sections.

5. The blind position of all nozzles associated with the equipment shall be strictly ensured and verified as per the approved P&ID and isolation requirements prior to commencement of maintenance activities.

6. Work Permit System:

   - Electrical Isolation Permits shall be mandatorily linked to the corresponding Work Permit through the permit management system, with the relevant Lock Out Tag Out (LOTO) details recorded in the Work Permit. For all high-risk vessel interventions involving vessel opening, confined space entry, or line breaking, the Work Permit shall not be authorized unless the corresponding LOTO certificate, latest gas test results, and decontamination records (including steaming, purging, and/or water washing logs, as applicable) have been verified.

   - The Issuing Authority and Permit Receiver, both shall ensure that the equipment has been effectively isolated and the requirements of the Lock Out Tag Out (LOTO) system have been complied, in accordance with Clause 6.3.2 of OISD-STD-105.

   - The relevant sections and conditions of the permits shall be appropriately ticked in the permit form. Poor communication can cause mistakes and accidents. Also, quantitative values and limits of gas test values shall be numerically specified in the work permits. All requirements as specified in Clause 6.4.2 of OISD-STD-105 shall be followed.

7. Personnel exposed to intense explosion should immediately be medically evaluated as blast related internal injuries may not manifest with immediate symptoms. After initial stabilization, as required, they should be shifted at the earliest to nearby hospitals/ trauma centers equipped with advanced diagnostic and treatment facilities.
8. CCTV cameras with video analytics, cognitive capabilities, artificial intelligence to be extensively used for real time monitoring of unsafe acts, unsafe conditions and safety violations as detailed in Clause 4.14.1 of Working Group report and Clause 7.6.8 of OISD-GDN-206. Adequate coverage of such cameras covering all risk-prone areas shall be ensured.

PHOTOGRAPHS:

Figure 1: Image showing blind position of steam line to first stage desalter

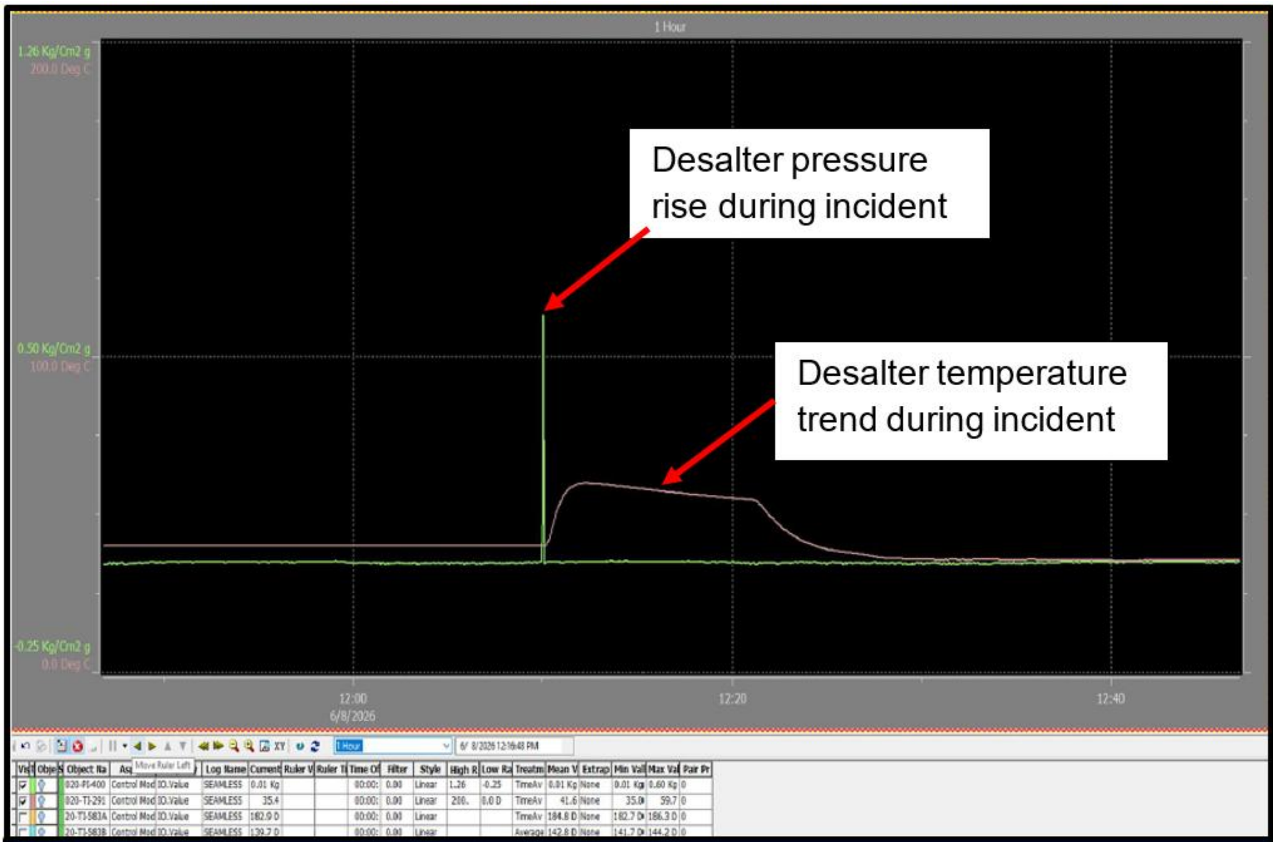

Figure 2: DCS trend of first stage desalter pressure and temperature at the time of incident

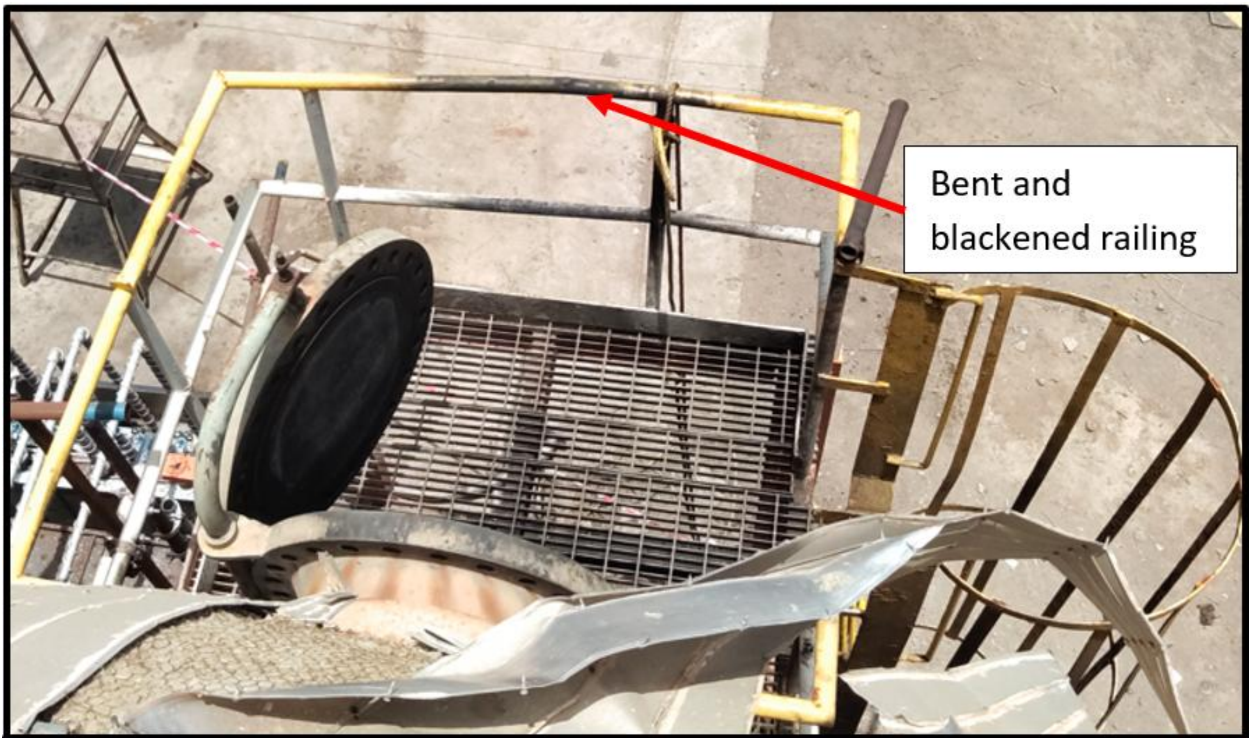

Figure 3: Incident Location Showing Blackened and Deformed Platform Railing

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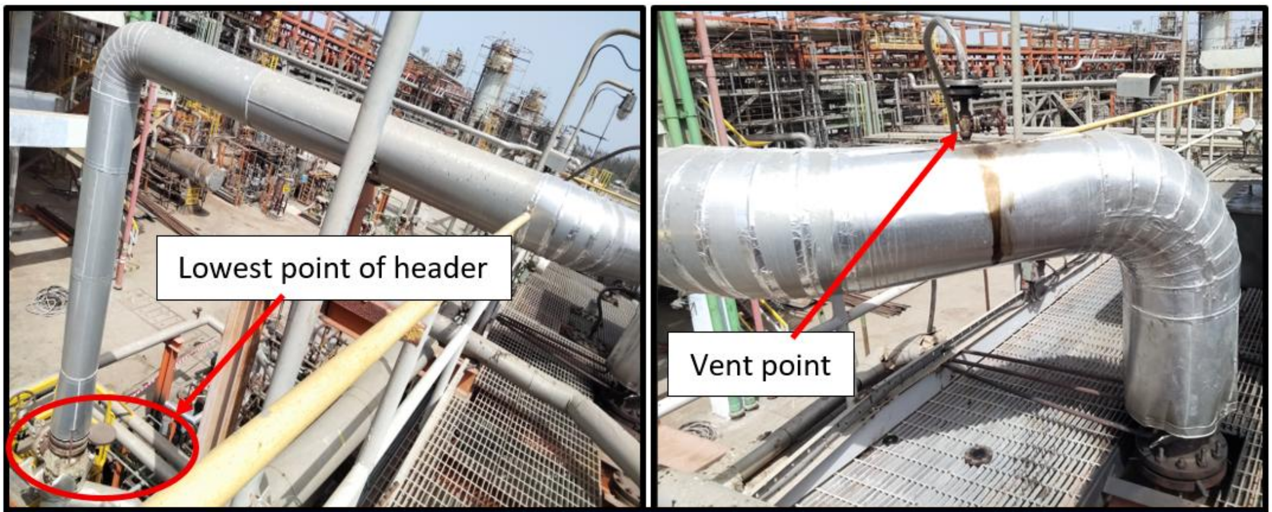

Figure 4: Desalter crude outlet line showing inverted U-shaped configuration without Low Point Drain (LPD) Facility

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